

Network Pruning Results on MNIST

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Running Network Pruning on device: CUDA

Dataset: MNIST

Model Parameters: 203,530

20% Pruning

Table 1: Pruning Results at 20% Sparsity

Method	Trial	Baseline Acc (%)	Finetuned Acc (%)	Acc Drop (%)
Spectral Edge	-	97.97	98.05	-0.08
Random Edge	trial0	97.98	98.04	-0.06
Random Edge	trial1	97.87	98.04	-0.17
Random Edge	trial2	97.96	98.25	-0.29
Random Edge	trial3	98.06	98.16	-0.10
Random Edge	trial4	98.06	98.10	-0.04
Random Edge	trial5	97.89	98.13	-0.24
Random Edge	trial6	98.03	98.04	-0.01
Random Edge	trial7	97.88	98.06	-0.18
Random Edge	trial8	98.00	98.04	-0.04
Random Edge	trial9	98.00	98.07	-0.07

30% Pruning

Table 2: Pruning Results at 30% Sparsity

Method	Trial	Baseline Acc (%)	Finetuned Acc (%)	Acc Drop (%)
Spectral Edge	-	97.97	97.93	0.04
Random Edge	trial0	97.94	98.07	-0.13
Random Edge	trial1	98.08	98.02	0.06
Random Edge	trial2	97.84	97.98	-0.14
Random Edge	trial3	97.93	98.05	-0.12
Random Edge	trial4	97.97	98.06	-0.09
Random Edge	trial5	97.98	98.13	-0.15
Random Edge	trial6	97.89	98.07	-0.18
Random Edge	trial7	97.91	98.08	-0.17
Random Edge	trial8	98.11	98.04	0.07
Random Edge	trial9	98.20	98.07	0.13

40% Pruning

Table 3: Pruning Results at 40% Sparsity

Method	Trial	Baseline Acc (%)	Finetuned Acc (%)	Acc Drop (%)
Spectral Edge	-	97.98	97.59	0.39
Random Edge	trial0	97.97	97.89	0.08
Random Edge	trial1	98.12	97.95	0.17
Random Edge	trial2	98.03	97.92	0.11
Random Edge	trial3	97.97	97.86	0.11
Random Edge	trial4	97.93	97.89	0.04
Random Edge	trial5	97.93	97.88	0.05
Random Edge	trial6	98.10	98.02	0.08
Random Edge	trial7	97.99	97.98	0.01
Random Edge	trial8	97.99	97.88	0.11
Random Edge	trial9	98.02	97.90	0.12

Overall Comparison

Table 4: Overall Pruning Comparison Results

Dataset	Prune Ratio	Method	Original Acc	Finetuned Acc	Acc Drop
MNIST	20%	random_edge	97.98	98.04	-0.06
MNIST	20%	random_edge	97.87	98.04	-0.17
MNIST	20%	random_edge	97.96	98.25	-0.29
MNIST	20%	random_edge	98.06	98.16	-0.10
MNIST	20%	random_edge	98.06	98.10	-0.04
MNIST	20%	random_edge	97.89	98.13	-0.24
MNIST	20%	random_edge	98.03	98.04	-0.01
MNIST	20%	random_edge	97.88	98.06	-0.18
MNIST	20%	random_edge	98.00	98.04	-0.04
MNIST	20%	random_edge	98.00	98.07	-0.07
MNIST	20%	spectral_edge	97.97	98.05	-0.08
MNIST	30%	random_edge	97.94	98.07	-0.13
MNIST	30%	random_edge	98.08	98.02	0.06
MNIST	30%	random_edge	97.84	97.98	-0.14
MNIST	30%	random_edge	97.93	98.05	-0.12
MNIST	30%	random_edge	97.97	98.06	-0.09
MNIST	30%	random_edge	97.98	98.13	-0.15
MNIST	30%	random_edge	97.89	98.07	-0.18
MNIST	30%	random_edge	97.91	98.08	-0.17
MNIST	30%	random_edge	98.11	98.04	0.07
MNIST	30%	random_edge	98.20	98.07	0.13
MNIST	30%	spectral_edge	97.97	97.93	0.04
MNIST	40%	random_edge	97.97	97.89	0.08
MNIST	40%	random_edge	98.12	97.95	0.17
MNIST	40%	random_edge	98.03	97.92	0.11
MNIST	40%	random_edge	97.97	97.86	0.11
MNIST	40%	random_edge	97.93	97.89	0.04
MNIST	40%	random_edge	97.93	97.88	0.05
MNIST	40%	random_edge	98.10	98.02	0.08
MNIST	40%	random_edge	97.99	97.98	0.01
MNIST	40%	random_edge	97.99	97.88	0.11
MNIST	40%	random_edge	98.02	97.90	0.12
MNIST	40%	spectral_edge	97.98	97.59	0.39